



BABA FARAH

EMBEDDED SYSTEMS & SOFTWARE ENGINEER

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With a background in Computer Engineering and Embedded Systems, I combine expertise in hardware architectures (C, VHDL, FPGA, microcontrollers) with software development. Experienced in physical data acquisition and connected solutions, I bring strong technical rigor, adaptability, and a problem-solving mindset to complex engineering challenges.

EDUCATION

- Master of Excellence in Computer Engineering and Embedded Systems, Ibn Zohr University – FSA, Agadir (2024–2026)
- Bachelor of Excellence in Computer Engineering and Embedded Systems, Ibn Zohr University – FSA, Agadir (2023–2024)
- Bachelor in Education – Computer Science, Higher School of Education and Training, Agadir (2020–2023)
- Baccalaureate in Physical Sciences (French option), Lycée Libres Agadir Ida Outanane (2022)
- Russian Language Diploma, Kharkiv Regional Institute of Public Administration, Ukraine (2020)
- Baccalaureate in Physical Sciences (Arabic option), Lycée Mohammed Derfoufi, Agadir (2019)

RECENT TECHNICAL PROJECTS

HealthTech Hackathon – Context-Aware Cardio-Motor Anomaly Monitoring | Dec. 2025

Designed a smart wearable bracelet for the acquisition and monitoring of cardio-motor data.

Tools: ESP32, MAX30102, MPU6050, IoT, Machine Learning, Flutter, Firebase

Autonomous Mobile Surveillance Robot | Nov. 2025

Designed an embedded robotic system capable of autonomous navigation and environmental monitoring.

Tools: Esp32, Esp32-CAM, motion sensors, camera, motors

AgriTech Hackathon – Plant Disease Detection | Dec. 2024

Developed an embedded IoT solution for plant disease detection based on environmental data and images.

Tools: Raspberry Pi 4, Camera, Soil Moisture Sensor, DHT22, Machine Learning, React

PROFESSIONAL EXPERIENCES

Embedded Systems Engineer – Final-Year Internship | OCP Jorf Lasfar | Feb. – Jun. 2026

Designed and developed an automated embedded system for physical data acquisition and processing.

Software Developer – Internship | MARSA MAROC, Agadir | Aug. – Sep. 2025

Developed a web application for SQL query management and execution using Laravel and MySQL.

Software Developer – Internship | National Ports Agency (ANP), Agadir | Jul. 2024

Developed a web application for IT asset management using Laravel and MySQL.

Software Developer – Final-Year Internship | ESEF Agadir | Feb. – May 2023

Developed an intelligent library management application in Java, including book, user, and loan management.

TECHNICAL SKILLS

- Programming Languages:** C, C++, Assembly, VHDL, Java, JavaScript, PHP, Python
- Web Development:** HTML5, CSS, PHP, Java, Python, WordPress
- Databases:** SQL, MySQL, SQL Server Management Studio, MongoDB
- Big Data / ETL:** Talend Open Studio, Hadoop, Spark, SSIS
- Development Tools and Platforms:** GitHub, Postman, Microsoft Office
- Modeling and Simulation:** UML, Merise, MATLAB, Quartus
- Operating Systems:** Windows, Linux

CERTIFICATIONS

- Technovate Hackathon AgriTech / HealthTech
- Automotive Engineering: Testing Strategies, UDS & Functional Safety (ISO 26262)
- Biomedical Visualisation – University of Glasgow

LANGUAGES

Arabic : Native ; **French :** Fluent ; **English :** Fluent ; **Russian :** Beginner